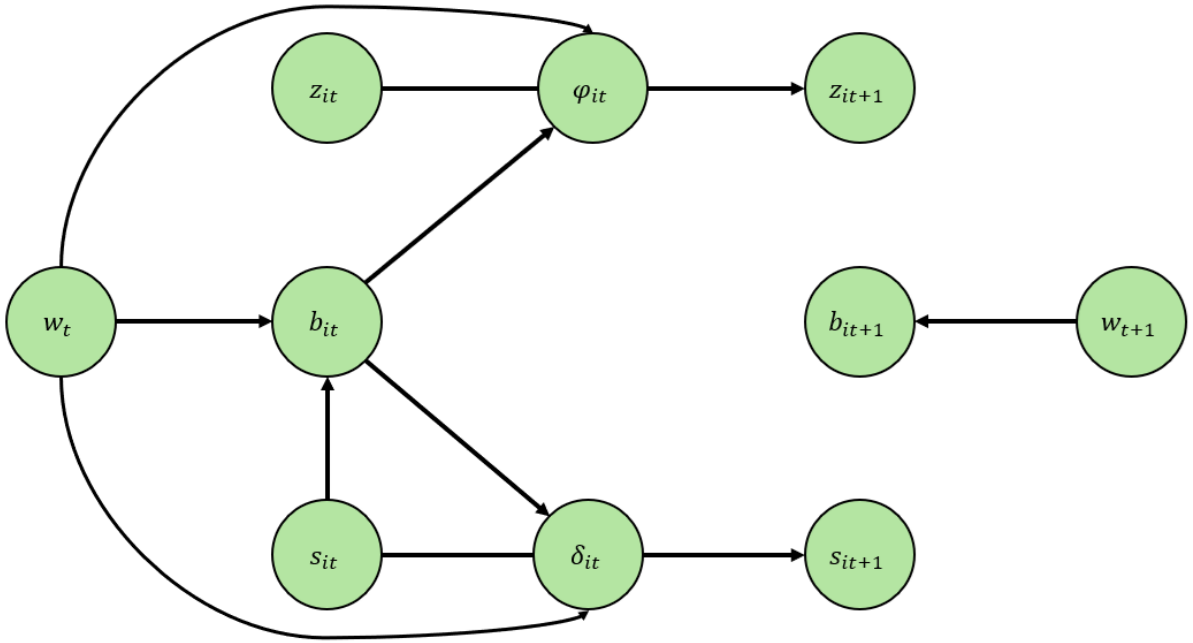
Introducing capture-recapture: cross-lags & life-histories







First, we will simulate our own little world…

12 foot alligator captured at Houston area shopping center

Nov 9, 2015 3:37 PM EST by David Chiu

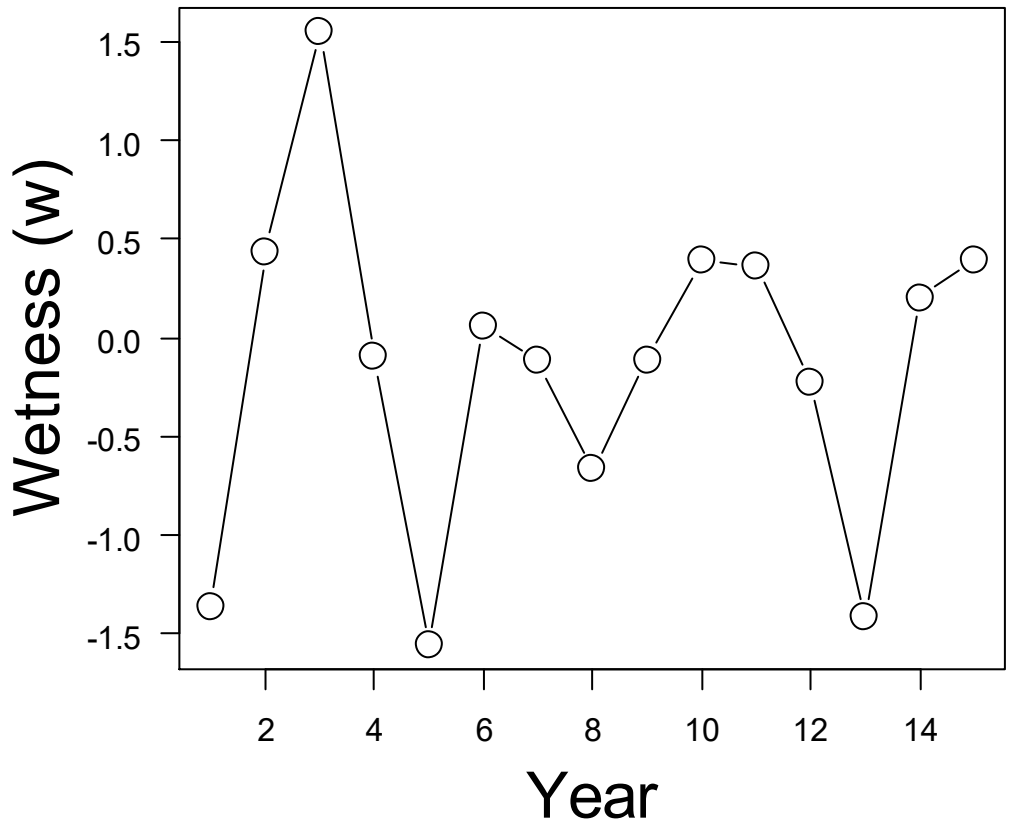


(ABC News) — A Texas shopping center had a surprise visitor Saturday morning.

A 12-foot alligator was discovered outside the First Colony Commons Shopping Center in Sugar Land, Texas, according to ABC station KTRK-TV. It weighed over 800 pounds.



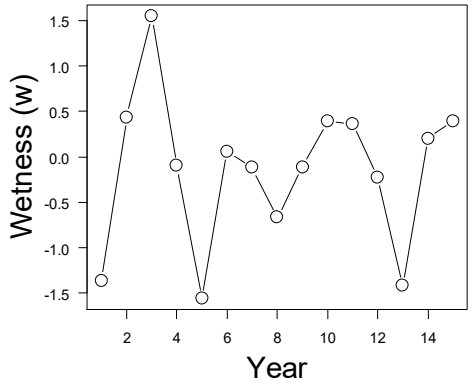
Nope, that was just Texas, we will simulate our own little world





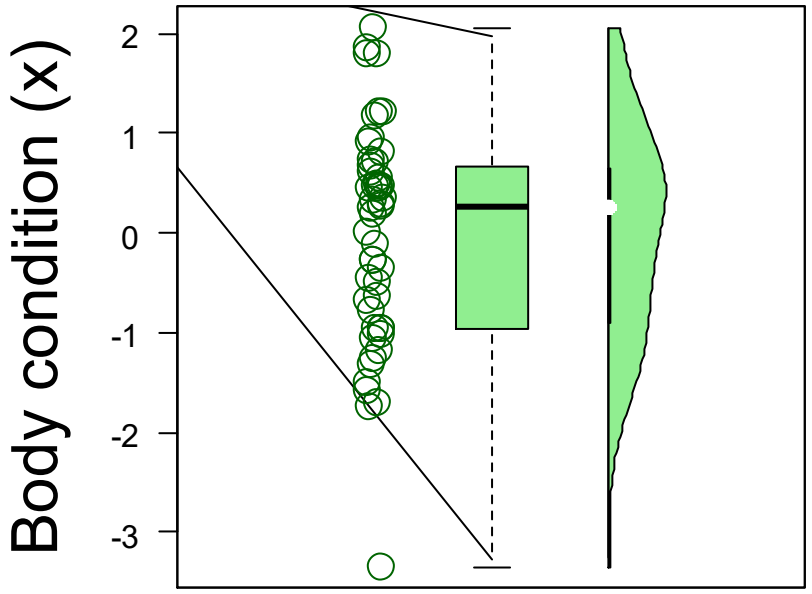
Wetland conditions will vary annually… zero Texans allowed

We'll fill our world with female alligators

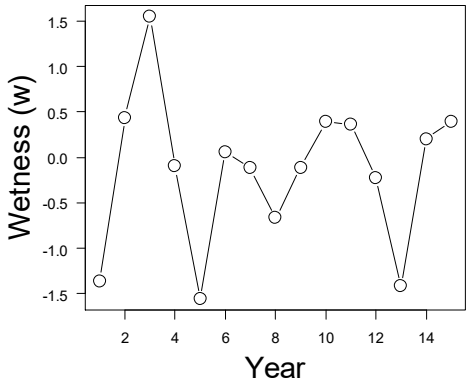


w_t

x_{it}

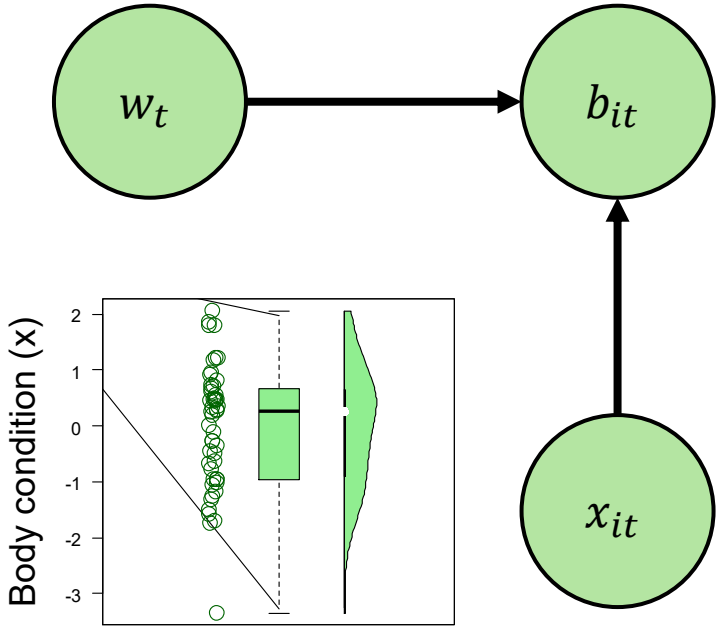


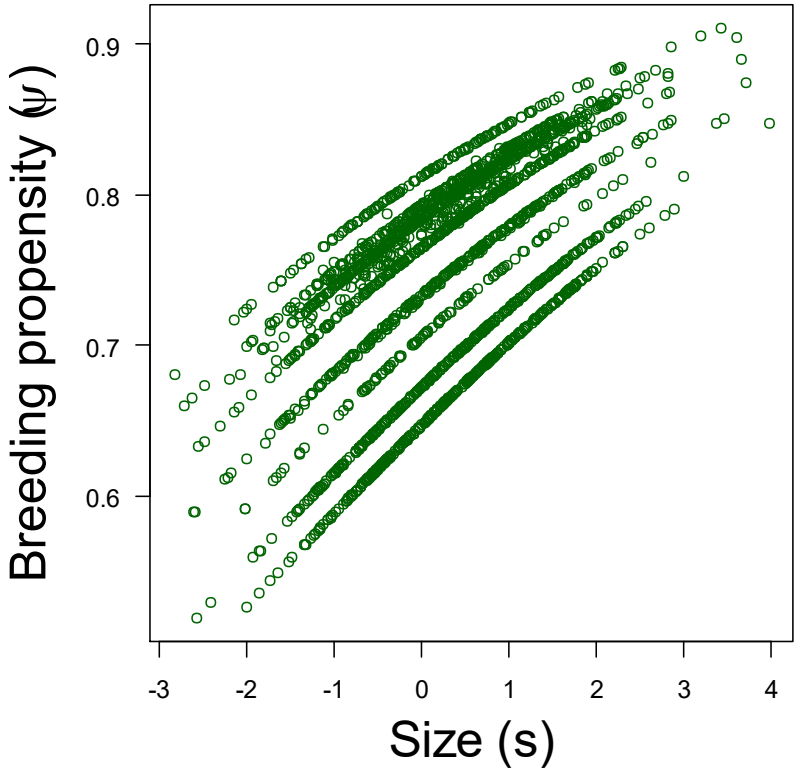
Alligators in better condition will breed (b) more often



$$\psi_{ij} = \text{logit}^{-1}(\beta_1 + \beta_2 x_{ij} + \beta_3 w_j)$$

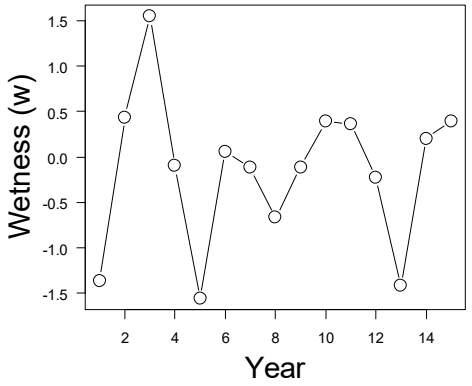
$$b_{ij} \sim \text{Bernoulli}(\psi_{ij})$$

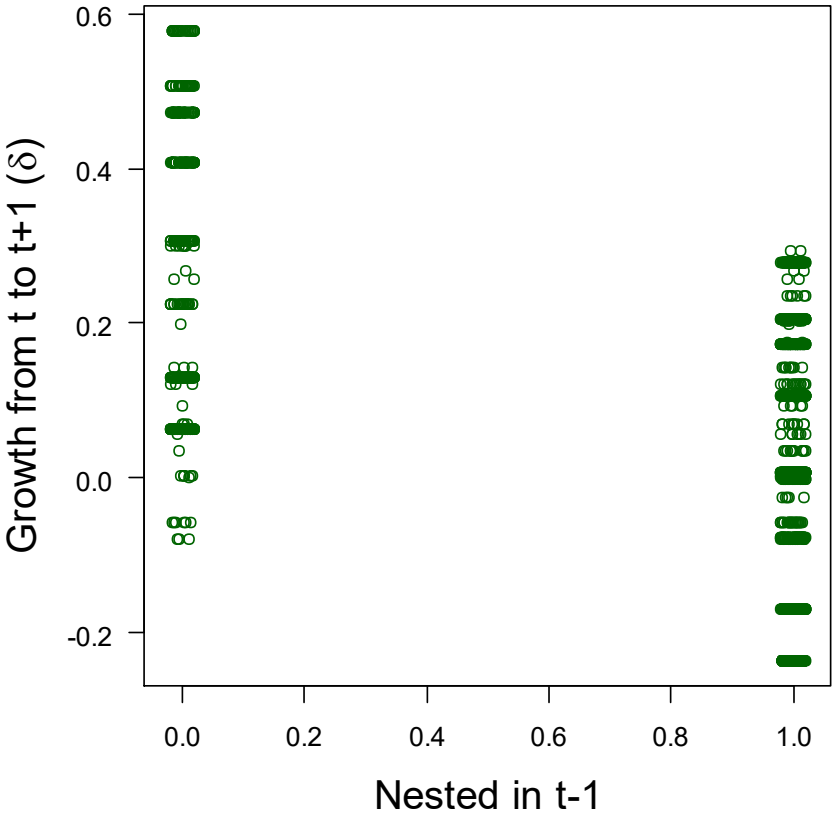


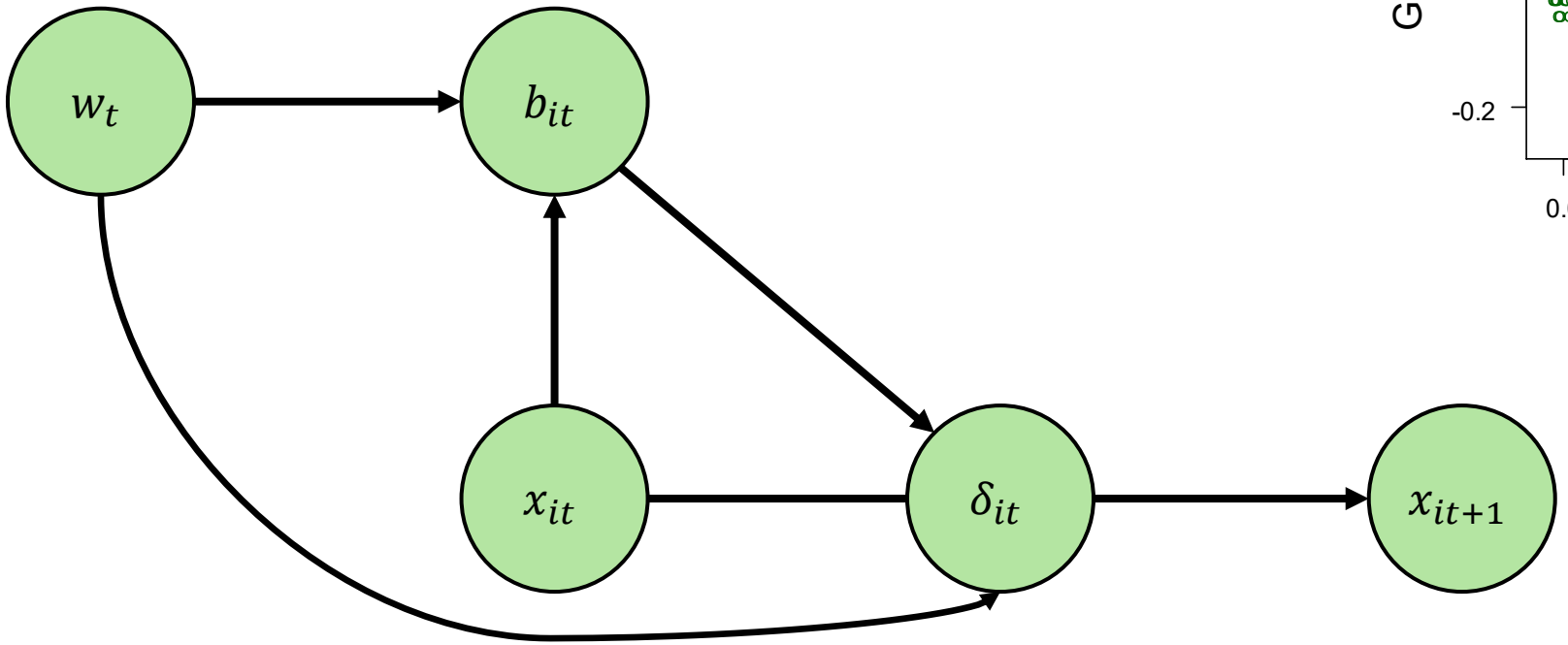


Wetter conditions = more breeding

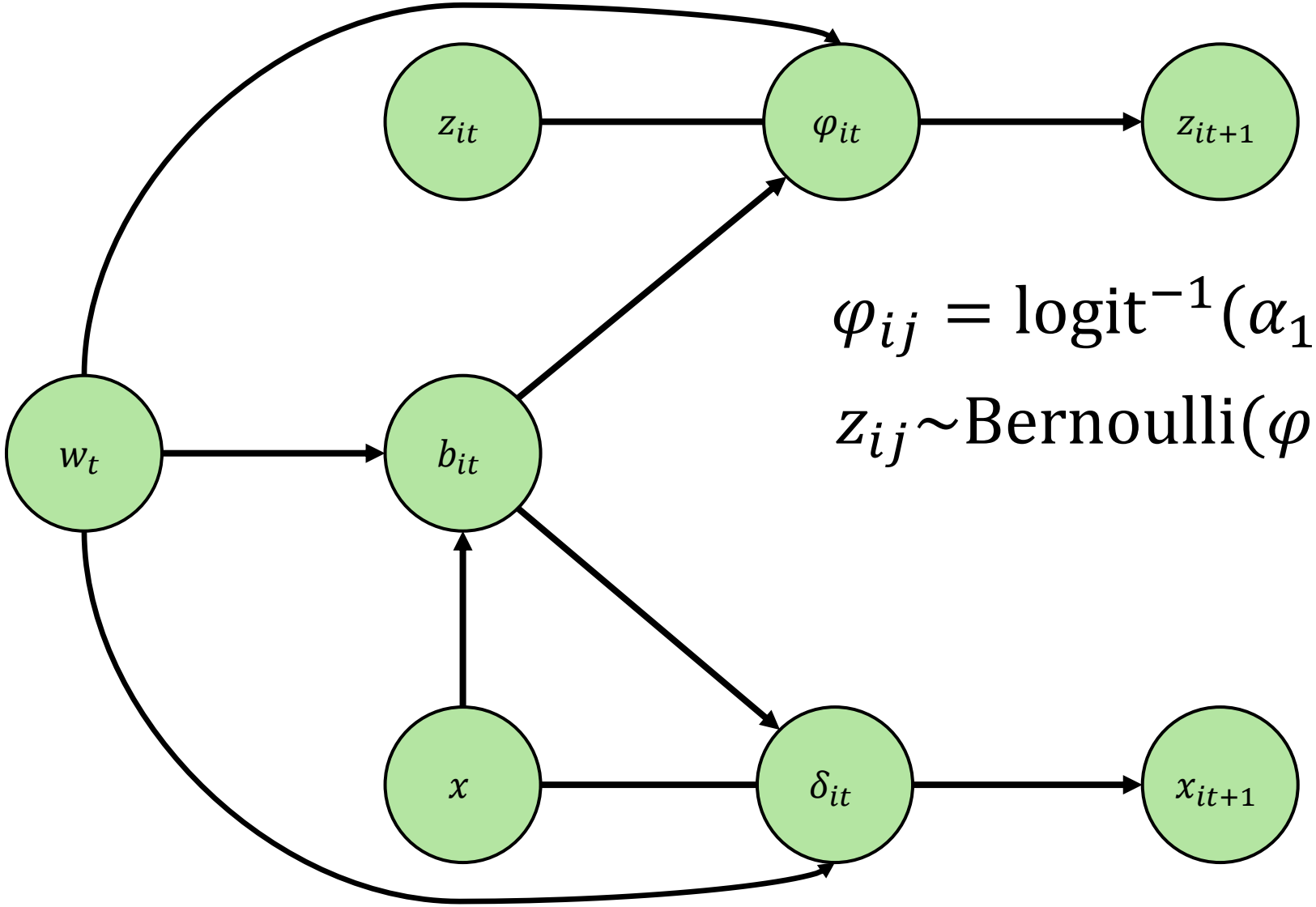
Alligators that breed will grow less





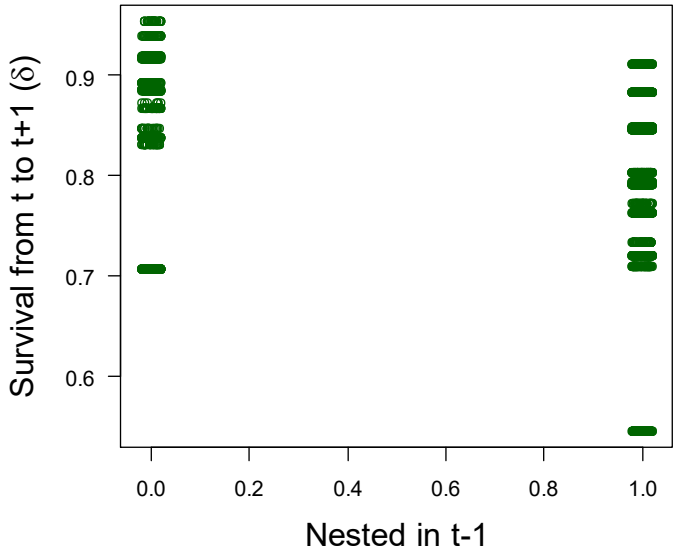


Alligators that breed will also be less likely to survive!

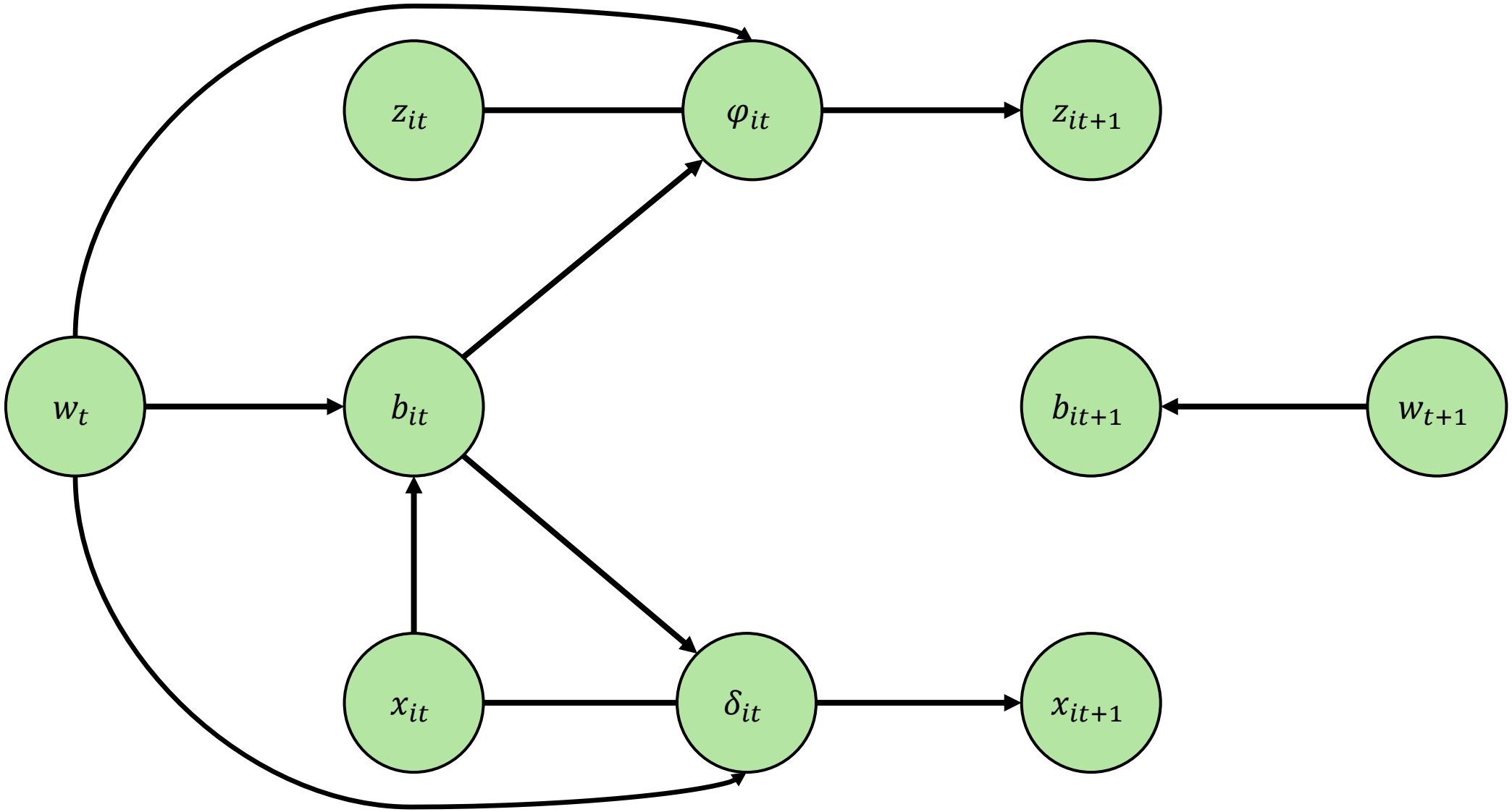


$$\varphi_{ij} = \text{logit}^{-1}(\alpha_1 + \alpha_2 b_{ij} + \alpha_3 w_j)$$

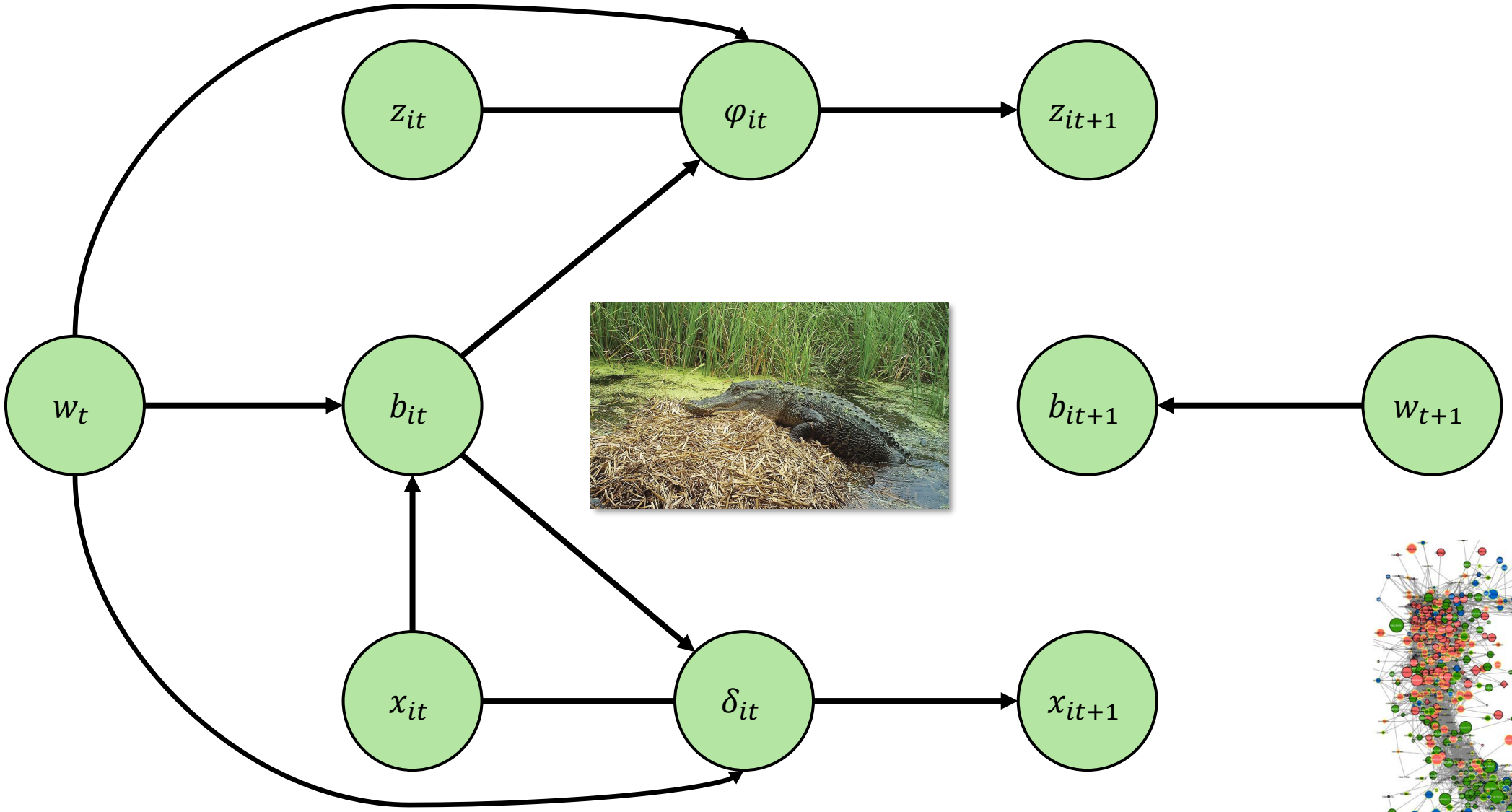
$$z_{ij} \sim \text{Bernoulli}(\varphi_{ij})$$

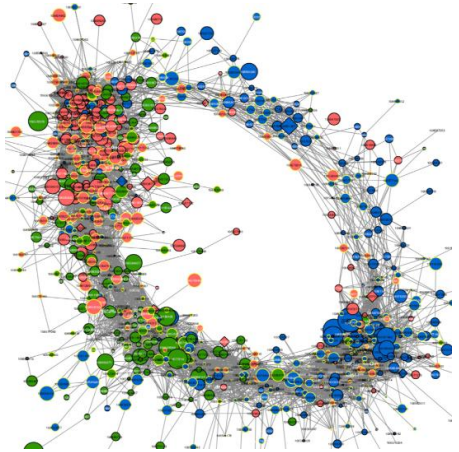


This will continue for a decade-and-a-half

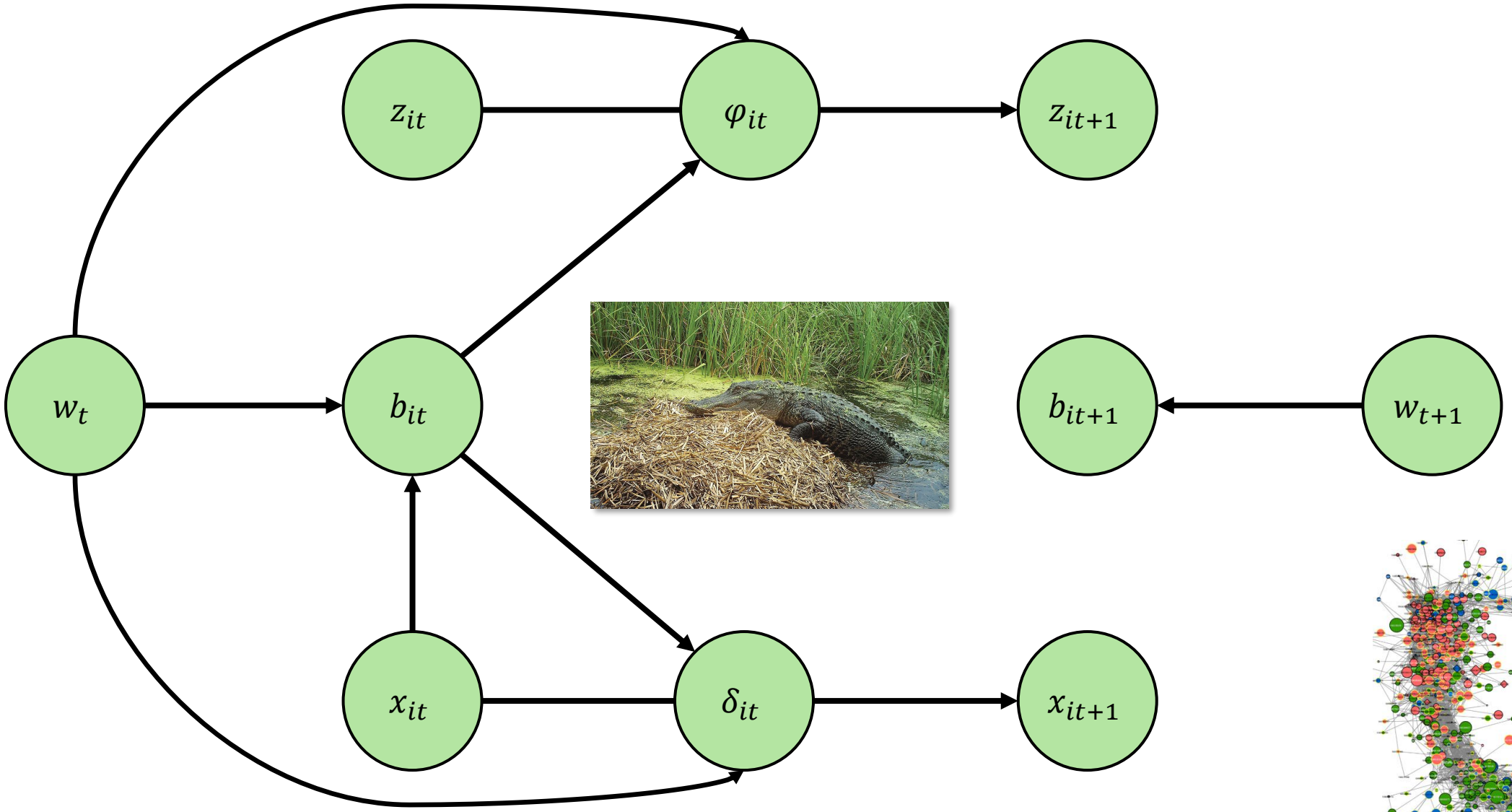


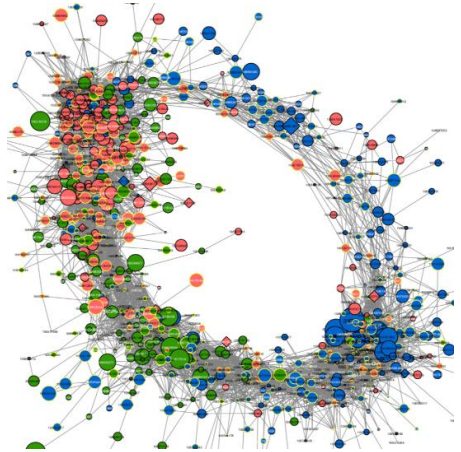
This looks complicated… is it?



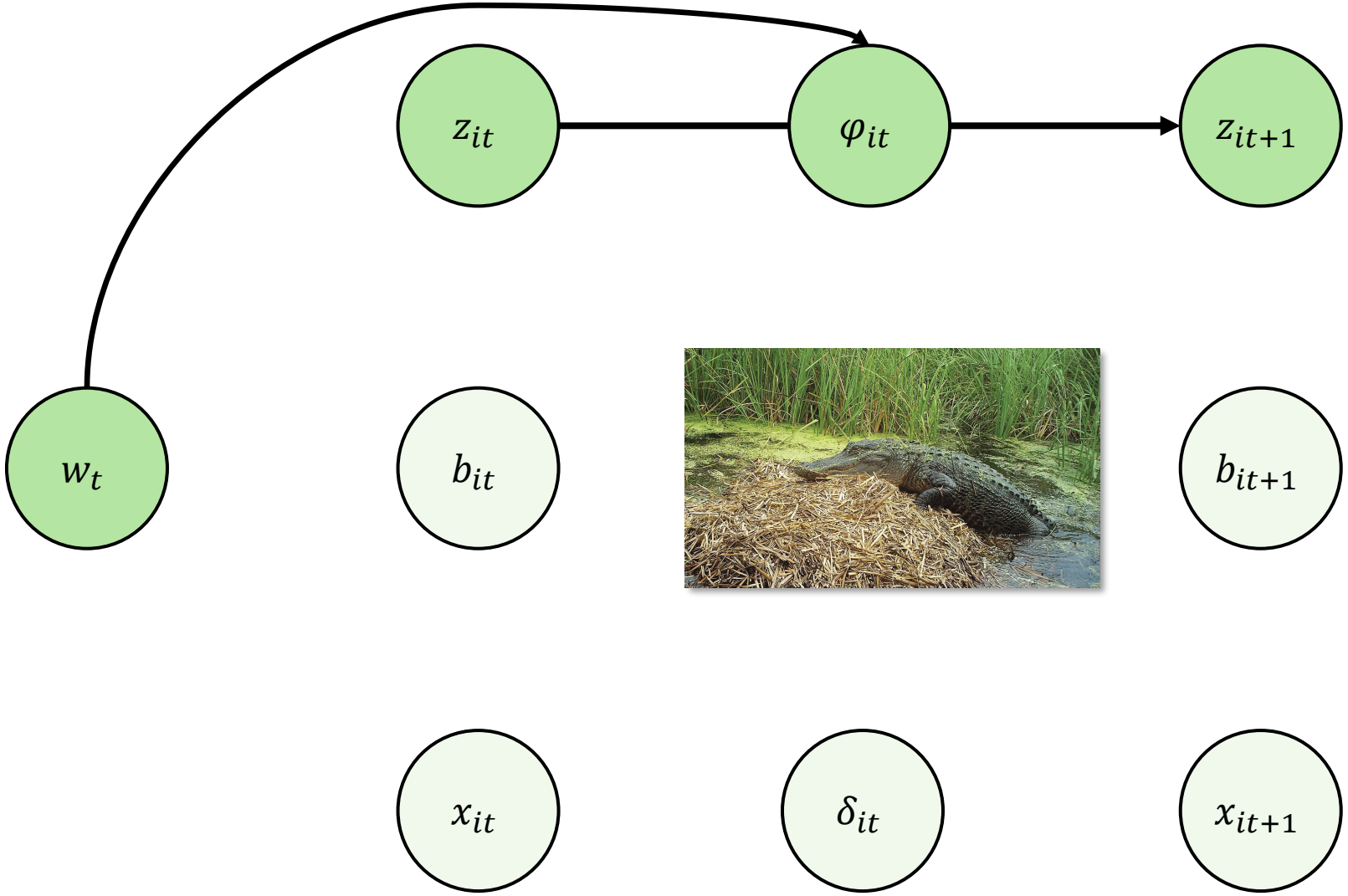


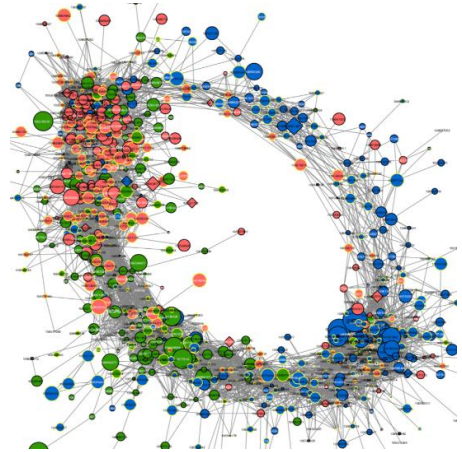
H: Animals are different, choices have consequences



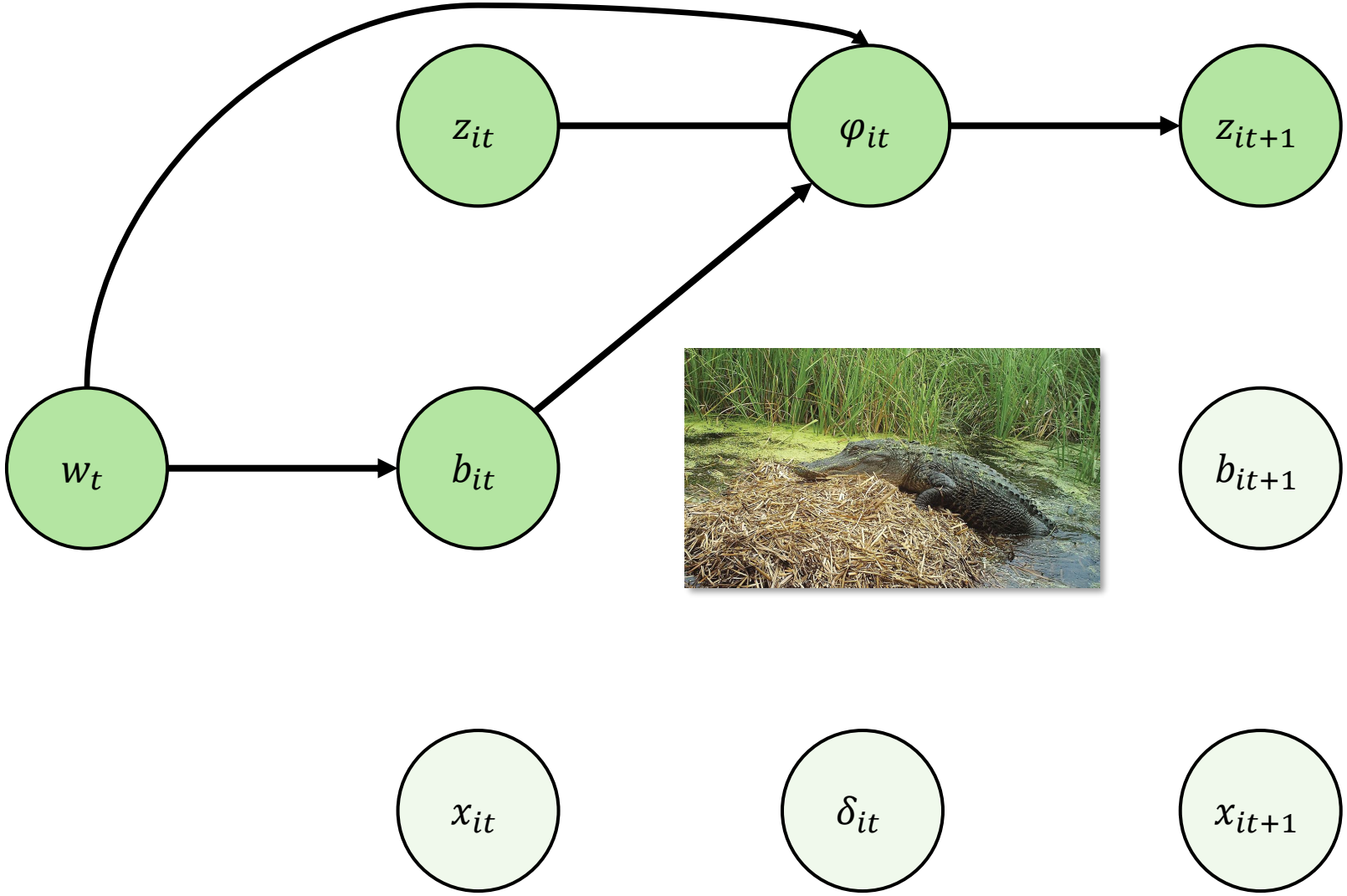


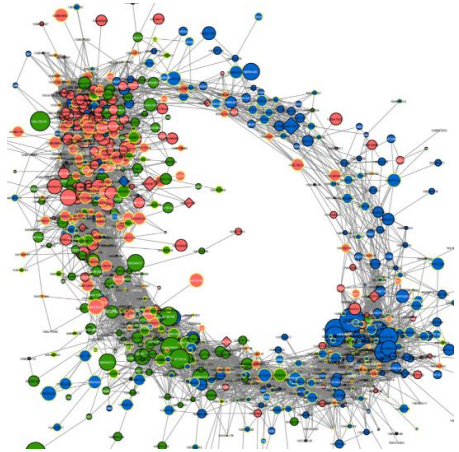
In m_1, we're only going to model survival (CJS)



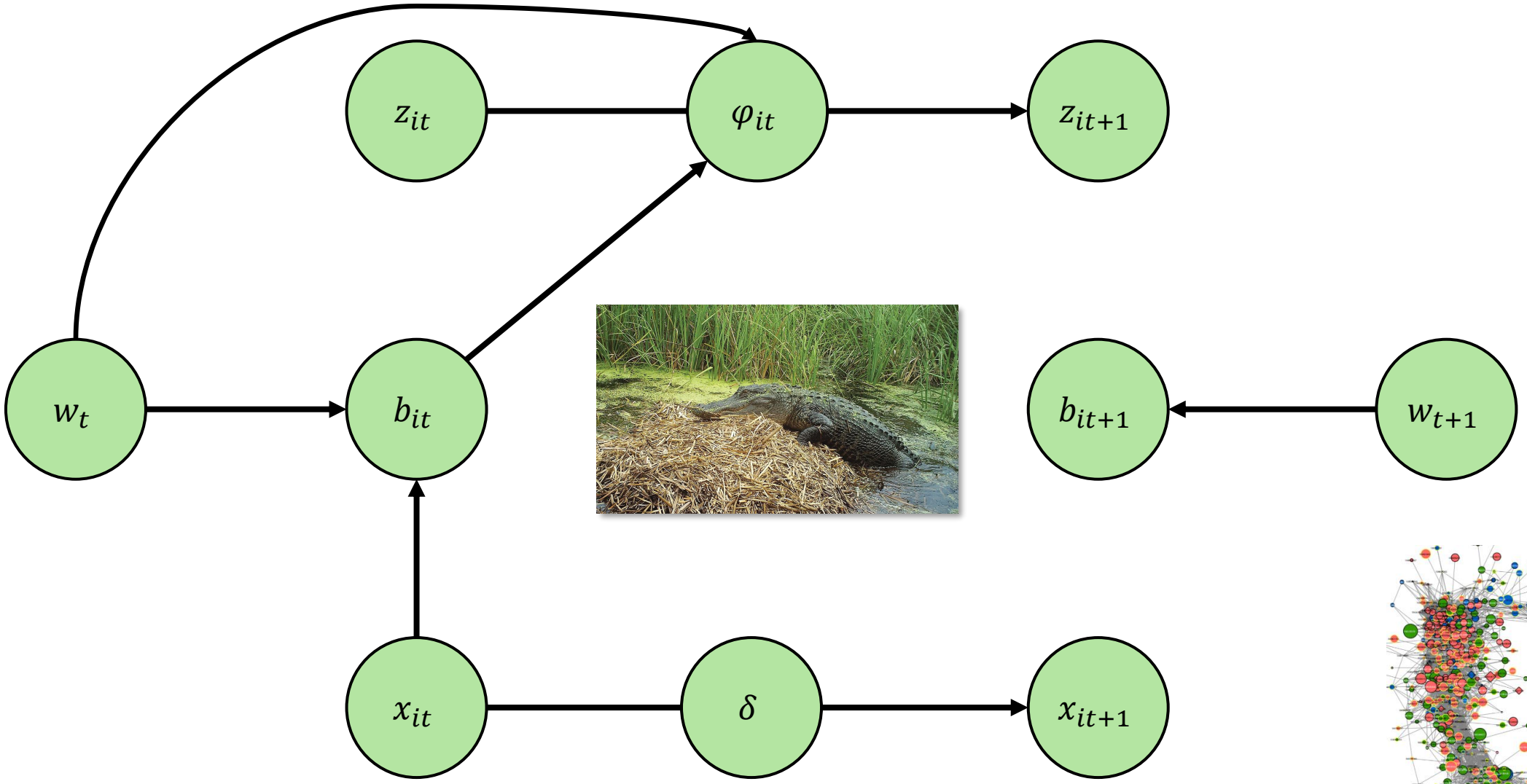


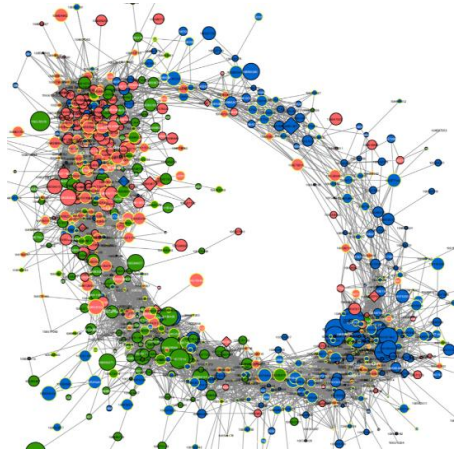
In m_2, we'll also incorporate breeding propensity (multistate)





In m_3, we'll partially incorporate growth (multistate)





Another wrinkle:
We'll observe marked alligators imperfectly

Let’s talk about this imperfect observation thing

(Hidden Markov models)

Imagine you release an animal and never see it again…

100000000000

What happened to that animal? How could we learn that?

What if there was a second animal?

100000000000

1111111111111

We released it and saw it every year thereafter

Imagine you release an animal and see it five years later…

100001

This gives us 10 pieces of information. What are they?

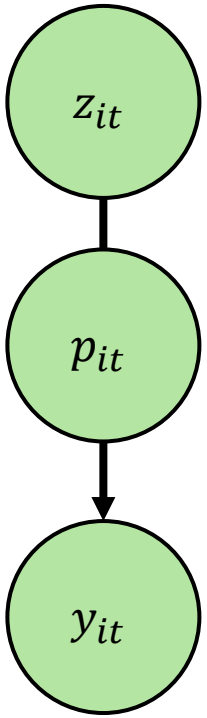
We can gather those pieces of information across many individuals

100001
111000
101010
111010
100000
100000
111110

And estimate those parameters

We can gather those pieces of information across many individuals

Ecological process



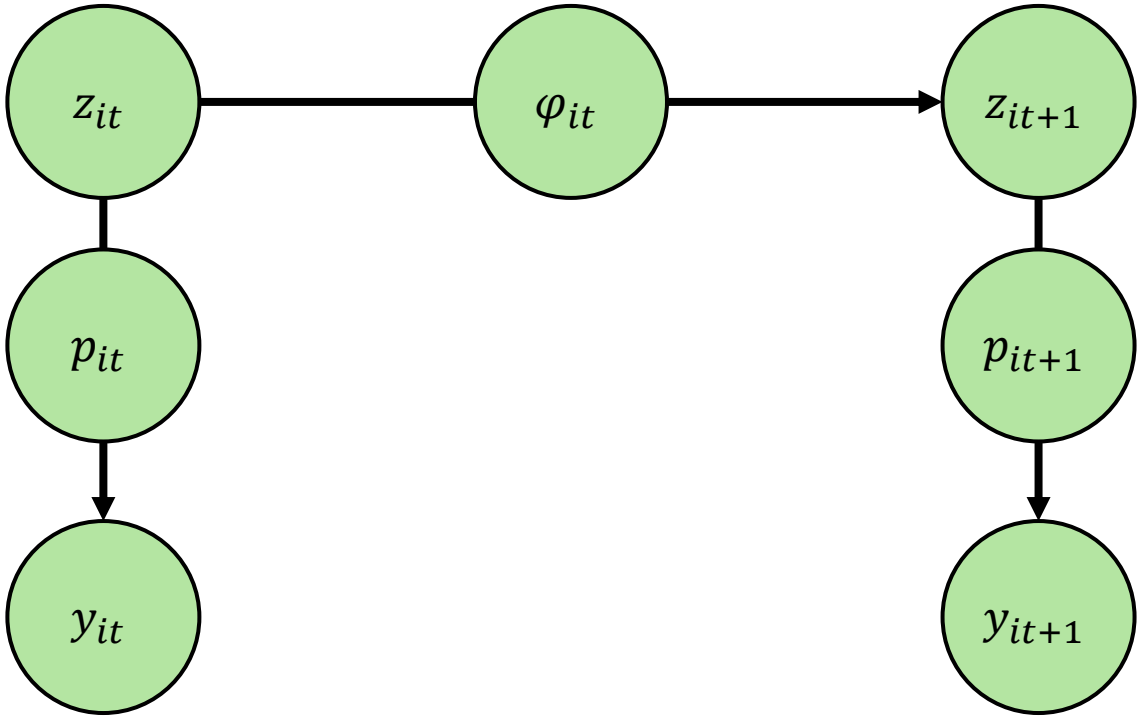
Observation process

$$y_{ij} \sim \text{Bernoulli}(z_{ij} \times p_{ij})$$

We can gather those pieces of information across many individuals

Ecological process

$$z_{ij} \sim \text{Bernoulli}(z_{ij-1} \times \varphi_{ij-1})$$



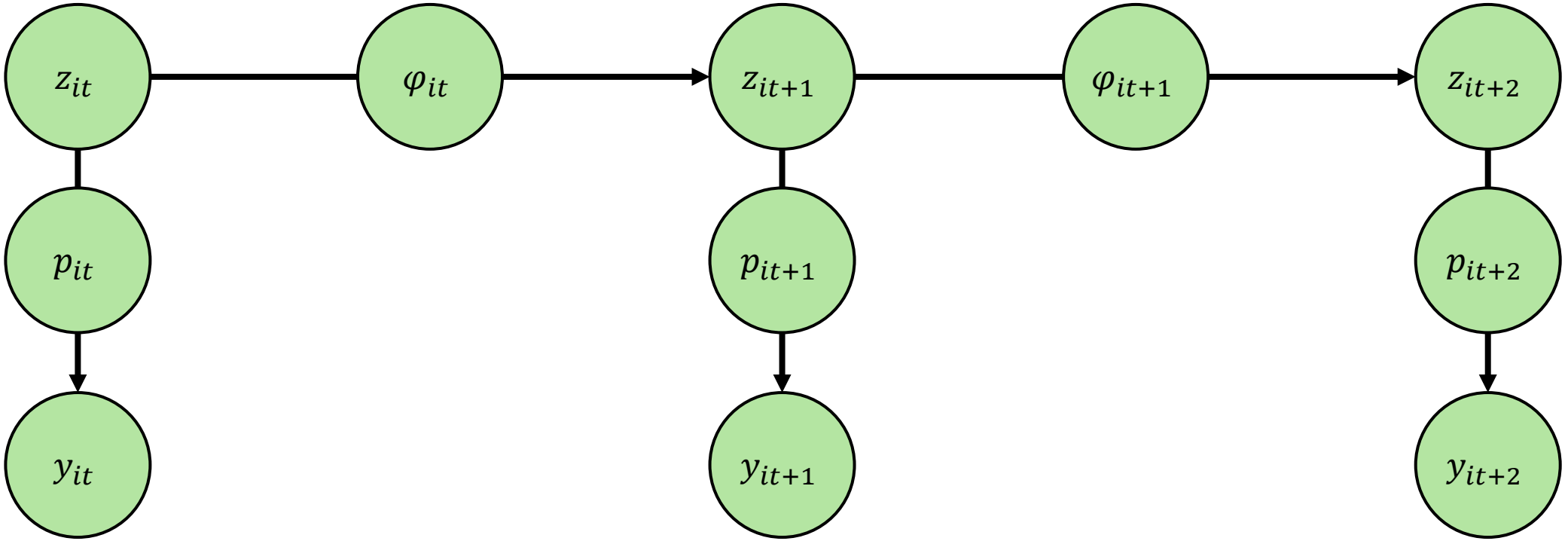
Observation process

$$y_{ij} \sim \text{Bernoulli}(z_{ij} \times p_{ij})$$

We can gather those pieces of information across many individuals

Ecological process

$$z_{ij} \sim \text{Bernoulli}(z_{ij-1} \times \varphi_{ij-1})$$



Observation process

$$y_{ij} \sim \text{Bernoulli}(z_{ij} \times p_{ij})$$

That’s the basic idea…

We can extend that idea in many ways, e.g., multistate models

To use multistate models…
we need to understand the categorical distribution

Bernoulli trials (2 outcomes, success or failure)

$$y_i \sim \text{Bernoulli}(\theta)$$
$$y_i \sim \text{Bernoulli}(0.5)$$



Data
y: heads (1)? or tails (0)?

Parameter
θ: the probability of heads

Categorical trials (2 or more outcomes)

$$y_i \sim \text{Categorical}([\theta, 1-\theta])$$
$$y_i \sim \text{Categorical}([0.5, 0.5])$$



Data

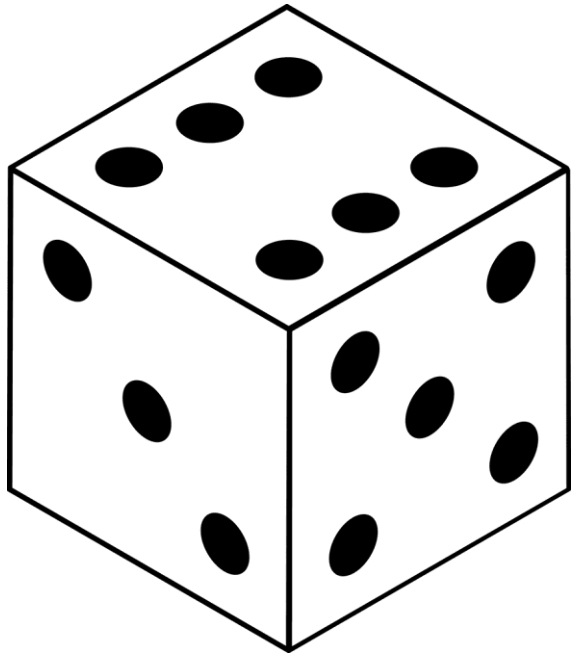
y: heads (1)? or tails (2)?

Parameter

θ: the probability of heads

Categorical trials

$$y_i \sim \text{Categorical}(\boldsymbol{\theta})$$
$$\theta = \left[\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right]$$



Categorical trials

$$y_i \sim \text{Categorical}(\boldsymbol{\theta})$$
$$\theta = \left[\frac{1}{20}, \cdots, \frac{1}{20}\right]$$

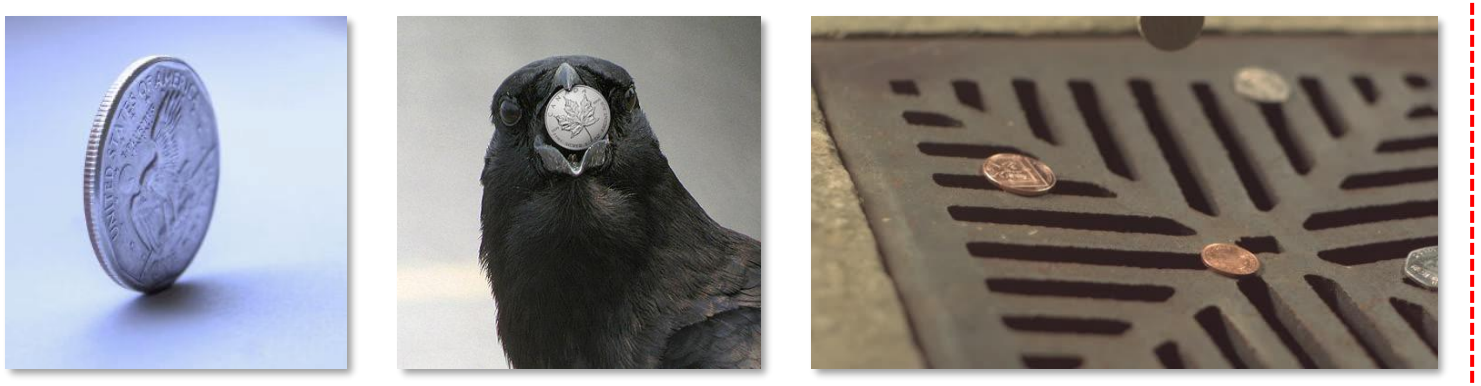


Categorical trials, outcomes can be a f(x) of >1 θ's

$$y_i \sim \text{Categorical}([\theta \times \delta, \quad (1-\theta)\delta, \quad 1-\delta])$$
$$y_i \sim \text{Categorical}([0.495, 0.495, 0.01])$$



Heads **Tails**



Dumb

Data/outcomes
y: heads (1), tails (2), or stupid (3)

Parameters
θ: the probability of heads
δ: probability 'not stupid'

Latent states

CJS

$$z_{i,t} \sim \text{Bernoulli}(z_{i,t-1} \times \varphi)$$

Multistate

$$z_{i,t} \sim \text{Categorical}(\Psi_{z_{i,t-1},i,t,})$$

Multistate capture histories

State = individual, categorical value that may change

Examples of states:

- Geographical locations (e.g., site 1 or site 2)
- Breeding status (e.g., breeder or non-breeder)
- Disease status (e.g., healthy or sick)

ID	1992	1993	1994	1995	1996	1997	1998	1999
1	0	1	2	2	0	2	0	0
2	1	1	1	0	0	0	0	0
3	0	2	0	0	0	0	0	0
4	0	0	2	1	0	2	0	0
5	0	0	0	1	1	2	1	1
6	0	0	0	2	2	1	2	0

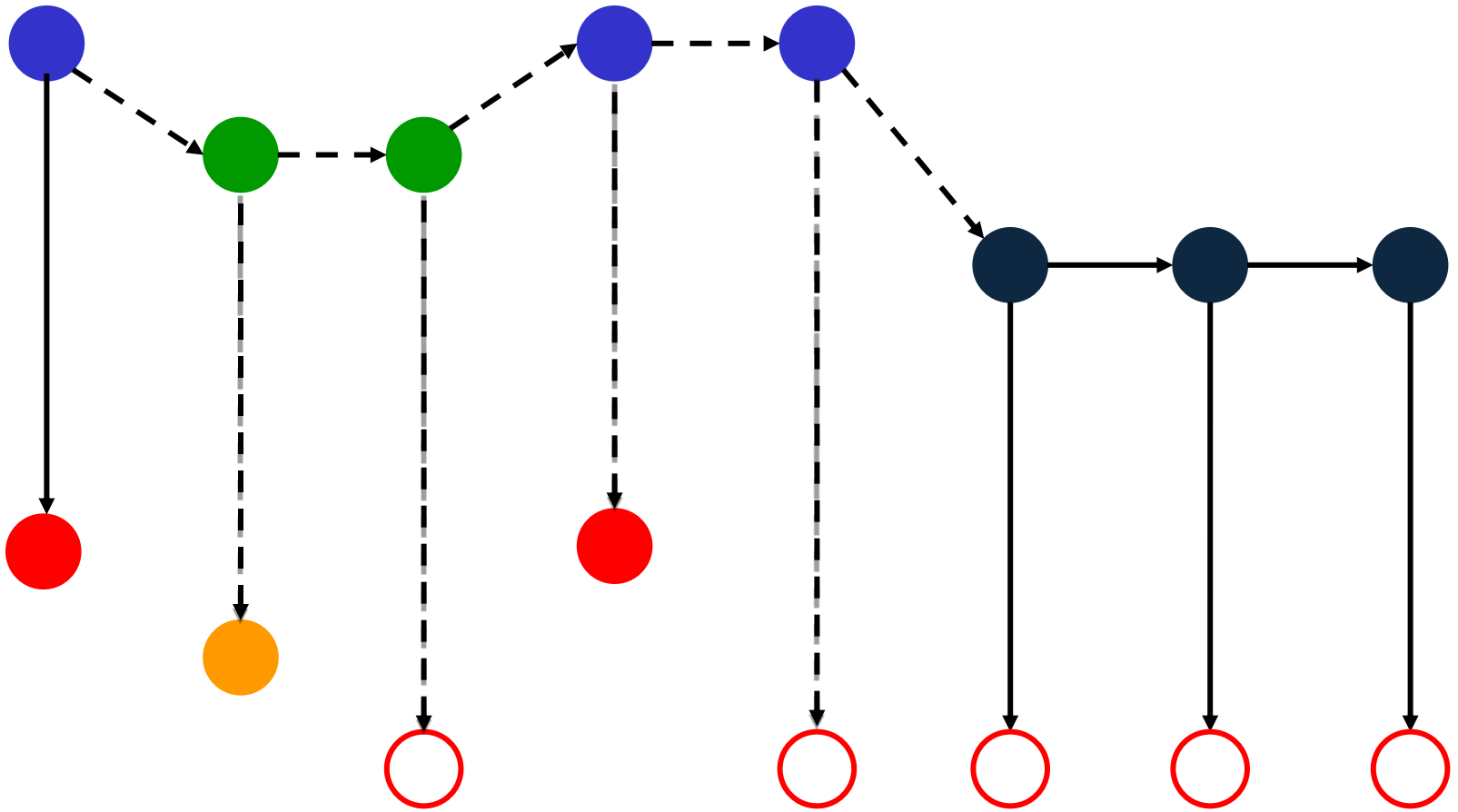
Conditional nature of the 2 processes (ch, [12010000])

State process

Site A

Site B

Dead



Seen at A

Seen at B

Not seen

Observation process

Stochastic process

Deterministic process

What you will be

What you are	*site A*	*site B*	*dead*
site A	Φ_{AA}	Φ_{AB}	$1-\Phi_{AA}-\Phi_{AB}$
site B	Φ_{BA}	Φ_{BB}	$1-\Phi_{BA}-\Phi_{BB}$
dead	0	0	1

How you're seen

What you are	*Seen at A*	*Seen at B*	*Not seen*
site A	p_A	0	$1-p_A$
site B	0	p_B	$1-p_B$
dead	0	0	1

State process

States at time t+1

States at time t	*site A*	*site B*	*dead*
site A	Φ_{AA}	Φ_{AB}	$1-\Phi_{AA}-\Phi_{AB}$
site B	Φ_{BA}	Φ_{BB}	$1-\Phi_{BA}-\Phi_{BB}$
dead	0	0	1

Observation process

Observations at time t

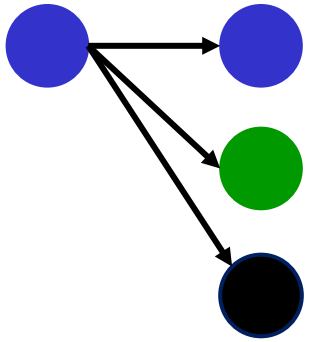
States at time t	*Seen at A*	*Seen at B*	*Not seen*
site A	p_A	0	$1-p_A$
site B	0	p_B	$1-p_B$
dead	0	0	1

State process

Site A

Site B

Dead



$$\Omega = \begin{array}{c} \\ \textit{States at time } t \\ \\ \end{array} \begin{array}{c} \textit{States at time } t+1 \\ \begin{array}{ccc} \textit{site A} & \textit{site B} & \textit{dead} \end{array} \\ \begin{bmatrix} \Phi_{AA} & \Phi_{AB} & 1-\Phi_{AA}-\Phi_{AB} \\ \Phi_{BA} & \Phi_{BB} & 1-\Phi_{BA}-\Phi_{BB} \\ 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{l} \\ \textit{site A} \\ \textit{site B} \\ \textit{dead} \end{array}$$

BUGS language:

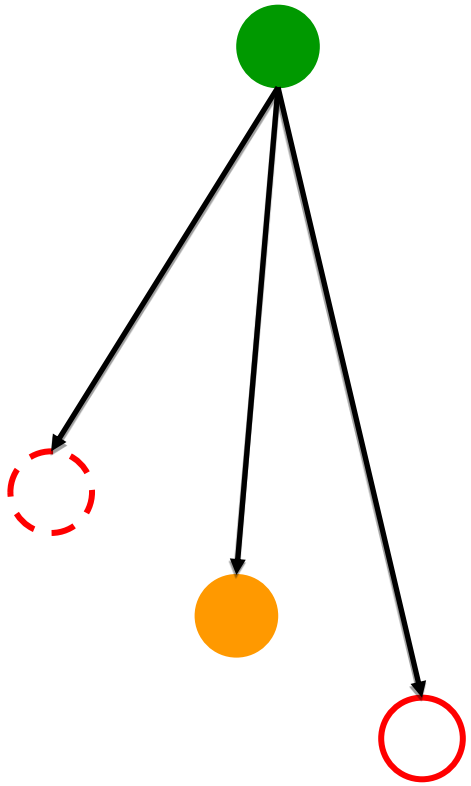
$$z_{i,t+1} : \; dcat\left(\Omega_{z_{i,t}},\right)$$

Observation process

Site A

Site B

Dead



Seen at A

Seen at B

Not seen

$$\Theta = \begin{array}{c} \\ \textit{site A} \\ \textit{site B} \\ \textit{dead} \end{array} \begin{array}{c} \begin{array}{ccc} \textit{Seen at A} & \textit{Seen at B} & \textit{Not seen} \end{array} \\ \begin{bmatrix} p_A & 0 & 1-p_A \\ 0 & p_B & 1-p_B \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

States at time t

Observations at time t

BUGS language:

$$y_{i,t} : \; dcat\left(\Theta_{z_{i,t},}\right)$$

Usual re-parameterisation

- $\Phi_{xy,t}$: probability to be in state *y* at time *t*+1, given presence in state *x* at time *t*

$$\begin{bmatrix} \Phi_{AA} & \Phi_{AB} & 1-\Phi_{AA}-\Phi_{AB} \\ \Phi_{BA} & \Phi_{BB} & 1-\Phi_{BA}-\Phi_{BB} \\ 0 & 0 & 1 \end{bmatrix}$$

- ϕ_x: probability to survive from time *t* to time *t*+1, given presence in state *x* at time *t*

- $\psi_{xy,t}$: probability to move from state *x* to state *y* before time *t*+1, given survival from time *t* to time *t*+1

$$\begin{bmatrix} \phi_A\left(1-\psi_{AB}\right) & \phi_A\psi_{AB} & 1-\phi_A \\ \phi_B\psi_{BA} & \phi_B\left(1-\psi_{BA}\right) & 1-\phi_B \\ 0 & 0 & 1 \end{bmatrix}$$

Now we need to simulate some capture-recapture data

$$y_{ij} \sim \begin{cases} \text{Bernoulli}(p_1) & z_{ij} = 1 \text{ and } b_{ij} = 1 \\ \text{Bernoulli}(p_2) & z_{ij} = 1 \text{ and } b_{ij} = 0 \\ 0 & \text{otherwise} \end{cases} \qquad p = [0.9, 0.5]$$





Specifying initial values (CJS)

Data

y = [1, 0, 0, 0, 0, 1, 0]

Known

z = [1, 1, 1, 1, 1, 1, NA]

We know it was alive, and can supply as data

Specifying initial values (MS)

Data

y = [1, 3, 3, 3, 3, 1, 3]

Known

z = [1, NA, NA, NA, NA, 1, NA]

Initial value(s)

z = [NA, 1|2, 1|2, 1|2, 1|2, NA, NA]

We know it was alive, but not breeding or non-breeding